Theory of Computation-PROBLEM SET 1

March 6, 2026

Due March 19th. Upload to learn.tsinghua.edu.cn.

Problem 1

Suppose that A and B are languages such that $A \circ B$ is regular. Moreover, suppose that B is regular. Prove or disprove: A is regular.

Problem 2

In general, for any string $w = w_1 w_2 ... w_n$, the reverse of w, written w^R, is defined as the string with the reverse order $w^R = w_n w_{n-1} w_{n-2} ... w_1$. For any language A, let $A^R = \{w^R : w \in A\}$. Show that A^R is regular, if A is regular.

Problem 3

If A is any language, let $A_{-\frac{1}{2}}$ be

$$A_{-\frac{1}{2}} = \{x : \text{for some string y}, |x| = |y|, yx \in A\}$$

Show that $A_{-\frac{1}{2}}$ is regular if A is regular.

Problem 4

In some cases, the pumping lemma for regular languages fails to show that a language is non-regular. Let $L = \left\{0^a 1^b 2^c : (a = 1 \wedge b = c) \vee a \neq 1\right\}$.

(a) Show that the pumping lemma holds for the language L (even though it isn't regular).

(b) Show that L is non-regular.

Problem 5 (Optional)

Prove that if f is primitive recursive then the finite sum $\sum_{y<z} f(x, y)$ and product $\prod_{y<z} f(x, y)$ are primitive recursive. (Note that the domain here is set of natural numbers $\mathbb{N}$. You can use conclusions in lectures.)